UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY

Document Prepared by Sekans Enerji

Project Title	Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Uzundere I 63.0 MW Hydroelectric Power Plant Project (hereafter referred to as the “Project” or “Uzundere HEPP”), Turkey, which is developed by KARADENİZ HES ELEKTRİK A.Ş. (hereafter referred to as the “project owner”) is a hydroelectric power plant in Rize province, Turkey. The project activity was started in 27/05/2010.

The Project has been implemented and operated by KARADENİZ HES ELEKTRİK A.Ş.. The project aims to generate electricity from hydro energy and feed it to the national electricity grid. The project was not rejected by another GHG program before. In addition, the project highly supports the sustainable economic development in the region.

The project will produce positive environmental and economic benefits through the following aspects:

- Displacing the electricity generated by fossil fuel fired power plants by utilising the renewable resources to avoid environmental pollution and GHG emissions,
- Contributing the economic development of the region by providing sustainable energy resources,
- Increasing the income and local standard of living by providing job opportunities for the local people,
- Reducing the blackout because of low voltage by lowering required capacity of the transformer.

The project was operational on 27/05/2010.

Implementation of the project will consist of construction of the following main items:

- Two diversion weirs, where water from the river is diverted into conveyance pipes,
- The conveyance tunnel, which is 11.324 km,
- The powerhouse with two vertical shaft Pelton turbines, each having acapacity of 31.6 MW.
- Two generators attacted to the facility, each with a power factor of 0,9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

Total installed capacity in the powerhouse is 63.068 MWm / 62.152 MWe.

The implementation of the project consists of two diversion weirs, where water from the river is diverted into conveyance pipes; a conveyance tunnel with a length of 11.324 m and powerhouse with two Pelton type vertical turbines. Each turbine has a capacity of 31.534 MWm / 31.076 MWe.

Uzundere HEPP is expected to generate 156,205 MWh electricity output per annum. The estimated amount of emission reductions due to the realization of the proposed clean energy project is 76,997tons CO_2 per annum.

The start date of the project activity is 27/05/2010[1].

Table 1 - Monitoring timeline of the Project Activity

Date (Period)	**Activity**
01/04/2010	Issuance of EIA Certificate
01/04/2010	Construction Start Date
27/05/2010	Commissioning of the Project
27/05/2010 - 01/12/2011	1st Monitoring Period
02/12/2011 - 02/12/2017	2nd Monitoring Period
27/05/2010-26/05/2020	First Crediting Period
27/05/2020-26/05/2030	Second Crediting Period

1.2 Sectoral Scope and Project Type

[1] Ministry Acceptance Protocol is available to the VVB.

According to UNFCCC sectoral scopes definition for CDM projects, the Project Activity is included in the Sectoral Scope 1, category “Energy industries (renewable - / nonrenewable sources)”.

1.3 Project Eligibility

Requirements document as below.

- The project applies methodology ACM0002 Version 20.0, which is an approved methodology under VCS.
- The project type is hydro and an eligible project type as per the 1.1. Eligible Project Types & Scope under Renewable Energy Activity Requirements.

  (a) Project shall generate and deliver energy services (e.g., mechanical work/electricity/heat) from non-fossil and renewable energy sources.

  (b) Project shall comprise of renewable energy generation units, such as photovoltaic, tidal/wave, wind, hydro, geothermal, waste to energy and renewable biomass.

- The project activity results in displacement of electricity from thermal power stations while contributing to sustainable development of Turkey. Hence, the project contributes to the VCS and Mission.
- Hydro is an approved project type.

1.4 Project Design

The project is a single green field investment and is not part of a project group or bundle.

Eligibility Criteria

- Type of project: Hydro
- Location of project: The project is located in Rize province, Turkey. Therefore, the project is eligible.
- Project Area, Boundary and Scale: The registered project activity is 63.0 MW as large scale.

1.5 Project Proponent

Organization name	KARADENİZ HES ELEKTRİK A.Ş.
Contact person	Orçun Altınbaş
Title	ARGE Engineer
Address	Fahrettin Kerim Gökay cad. No:36 Altunizade 34662 Üsküdar – İstanbul - Turkey
Telephone	0090-216-5442400
Email	orcun.altinbas@iltekenerji.com.tr

1.6 Other Entities Involved in the Project

Organization name	Sekans Enerji Ltd. Şti.
Role in the project	Consultant
Contact person	Sıla Duran
Title	Consultant
Address	Eminey Evleri mah. Eski Büyükdere C. No:1/1 Sapphire AVM B:04 Kağıthane/İstanbul
Telephone	-
Email	sila@sekansdanismanlik.com

1.7 Ownership

KARADENİZ HES ELEKTRİK A.Ş.

1.8 Project Start Date

The project start date is 27/05/2010[2]

1.9 Project Period

Start date of the first crediting period: 27/05/2010

End date of the first crediting period: 26/05/2020

[2] Ministry Acceptance Protocol

Start date of the second crediting period: 27/05/2020

End date of the second crediting period: 26/05/2030

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
27.05.2020-31.12.2020	46,198
2021	76,997
2022	76,997
2023	76,997
2024	76,997
2025	76,997
2026	76,997
2027	76,997
2028	76,997
2029	76,997
01.01.2030-26.05.2030	30,798
Total estimated ERs	769,969
Total number of crediting years	10
Average annual ERs	76,997

1.11 Description of the Project Activity

The Uzundere I 63.0 MW Hydroelectric Power Project entails the construction and operation of a 63 MW hydroelectric power plant, the construction of a transmission line, connection canals, concrete tunnels and penstock. The project involves two vertical shaft Pelton turbines, each having a capacity of 31.6 MW. The first-hand turbines which are used in the project are supplied from Alstom Power Systems Hydro-Iberica. The turbines are Pelton type with vertical axis.

The net electricity production (delivered to the grid after losses and consumption in the plant) from the plant is estimated to be 156,205 MWh per annum. The amount of electricity generated by the project is not influenced by factors outside the project boundary such as other power plants or demand for electricity. Regarding the actual operation of the project activity, the first temporary acceptance protocol signed by the Ministry of Energy and Natural Resources is dated 27/05/2010 for the commissioning of one of the two vertical shaft Pelton turbine in the project. The second temporary ministry acceptance protocol is dated 07/11/2010 for the other turbine.

The baseline scenario is the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid-connected power plants and by the addition of new generation sources. The Baseline Scenario is detailed in section 3.4. In the project scenario, the project generates electricity from hydro power and result in emission reductions in parallel with its electricity generation values. Since Turkey's grid mainly consists of thermal power plants, this would have resulted in GHG emissions. Briefly, in the absence of the project activity, the electrical energy would have been delivered to the grid through a mix of existing power generation resources, as described in more detail in section 3.4.

In addition to displacing the electricity generated by fossil fuel fired power plants by utilising the renewable resources to avoid environmental pollution and GHG emissions, the project activity has increased the income and local standard of living by providing job opportunities for the local people and contributed the economic development of the region by providing sustainable energy resources.

1.12 Project Location

The project location is in the Black Sea Region of Turkey, within the province of Rize. The project is on the Uzundere Creek. The altitude is 817.59 m at powerhouse location. The project's geographical location is 40º50´16´´N - 40º54´04´´N and 40º45´4´´E - 40º49´42´´E.

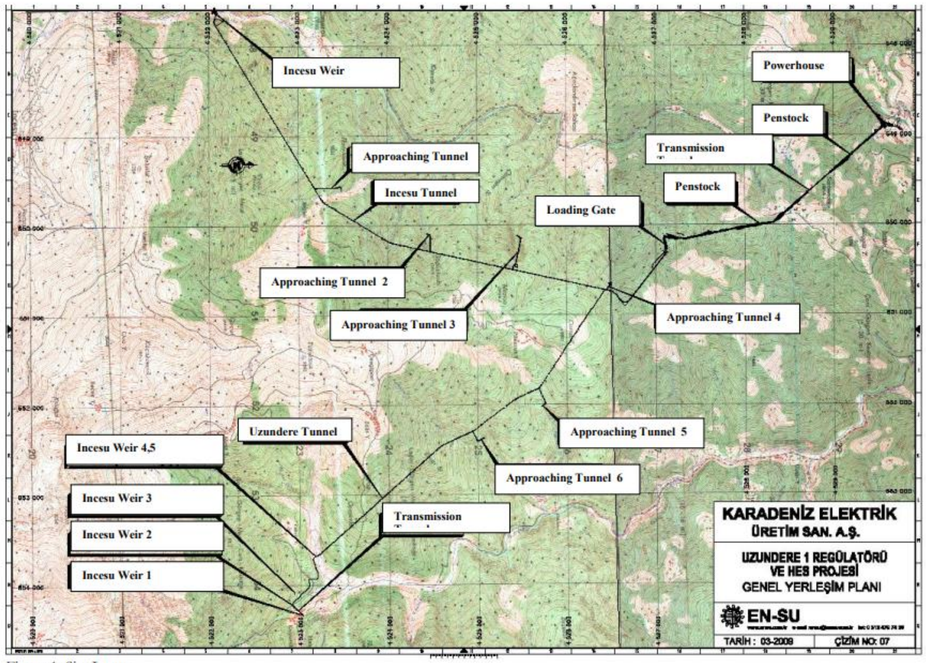

Figure 1. Project Site Layout

Figure 2. Project location on Turkey Map

1.13 Conditions Prior to Project Initiation

Please see Section 3.4.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

Table 2 - Relevant laws and regulations

Relevant Laws	Number/ Enactment Date	Aim and Scope
Environmental Law *Environmental Impact Assessment Regulation	Nr. 2872 / 17/07/2008	The approval is requested for power plants from Ministry of Environment and Forest as Electricity Licence Regulation requests project to be in line with the environmental law.

Electricity Market Law *ElectricityLicence Regulation *Electricity Market Balancing andConciliation Regulation	Nr. 4628 / 03/03/2001	Regulating procedures of electricity generation, transmission, distribution, wholesale, retail for legal entities. Two regulations issued under the law; one for generation licence and the other for market price balancing and conciliation.
Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electrical Energy	Nr. 5346 / 18/05/2005	Aims to extend the utilization of renewable energy for electricity generation and identifies method and principles for power generation from renewable resources in an economical and conservative manner as well as certification of the electricity generated from renewable resources.
Energy Efficiency Law	Nr. 5627 / 02/05/2007	Identifies method and principles for industry, power plants, residential buildings and transport to imply necessary measures for energy efficiency during electricity generation, transmission, distributionand consumption.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has never been included in an emissions trading program or any other binding limits. It has neither received any kind of environmental credits nor been registered under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project was not rejected by any other GHG program before.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

N/A

1.16.2 Other Forms of Environmental Credit

There is not another form of environmental credit generated by the project.

1.17 Additional Information Relevant to the Project

Leakage Management

N/A

Commercially Sensitive Information

N/A

Sustainable Development

The project produces electricity from renewable energy sources using hydro as the power source and to contribute to Turkey's growing electricity demand through a sustainable and low carbon technology. The project displaces the same amount of electricity generated by the grid dominated with fossil fired power plants. The project is expected to generate 156,205 MWh annually.

The annual emission reduction estimated by the project is 76,997tonnes CO2eq, approximately. While this number of emissions is mitigated, technology transfer is also realized as benefitting from hydropower.

The project contributes to improve the environmental situation in the region and in the country as avoiding fossil fuel-based electricity will enhance the air quality and help to reduce the negative effects on the climate. Through renewable technologies and hydro-based electricity sustainable and climate friendly development is promoted.

During construction and operational period, the project has created employment opportunities for the local community. The project contributes the economic development of the region by providing sustainable energy resources. The project provides workers with a safe and healthy work environment and is not complicit in exposing workers to unsafe or unhealthy work environments.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

There is not any negative environmental or socio-economic impact.

2.2 Local Stakeholder Consultation

Although the domestic laws and regulations (Regulation on Environmental Impact Assessment (Date:16.12.2003, Number: 25318), and Regulation on Environmental Impact Assessment (Date: 17.07.2008, Number: 26939). According to the Article 7 of both) do not require any stakeholders' participation, both in the project planning and construction phase, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings.

In these meetings authorized people from Project Proponent introduced project to the local people in terms of the size of the project, expected length of the construction period and to give details about how this project will impact their lives. The information given mostly focused on how the project might have some environmental effects and how these issues will be mitigated by the investor. These meetings were finalized in November 2007 (These meetings were the ones held before the Rize Administrative Court and the Court has ordered the preparation of an EIA.

The feedback from the stakeholders was reflected on the project design and implementation. Also, a contact person from Project Proponent and relevant contact information was announced in the relevant meetings for ongoing communication.

The project owner used different channels to invite comments by stakeholders. Most of the meetings were held through the Mukhtar (village governors) and Council of Elderly (Ihtiyar Heyeti; the village council) of Cataldere village, and the officials of Municipality of Kaptanpasa. The project owner visited various households in the project area to comprehend their needs and to identify how the project could assist in meeting the social and economic needs of the community.

The contact information of the plant responsible exist at the Mukhtars of Çataldere and İncesu villages, the project owner and local community are always in touch. The project owner regularly checks with the Mukhtar if any complaint or a request exists. Any complaint or need from the local community could directly be received by the project owner and appropriate contributions or improvements are made to the local community. Regarding the monitoring period, signed letters by the Mukhtars of Çataldere and İncesu (dated on 26/02/2021) have been provided as declaring that the related information has been available to the villagers and they did not have any complaint.

There is no update or any change to the project design after the registration of the project.

The online site visit for the renewal crediting period with DOE was made on 18/04/2022.The local people were interviewed and the general outcome of the interviews was positive verbally.

2.3 Environmental Impact

The specific benefits of the project are:

- reduce greenhouse gas emissions in Turkey compared to the business-as-usual scenario,
- help to stimulate private sector participation in hydro power industry in Turkey,
- create employment during the construction and the operation phase of the plant,
- relatively reduce some other pollutants from power generation industry in Turkey, compared to a business-as-usual scenario,
- help to diminish Turkey's increasing energy deficit,
- diversify the electricity generation portfolio and reduce dependency on import of other energy sources.

There is no endangered fauna and flora in the region. The project design preferred glass fiber reinforced polyester pipes rather than open canals where necessary to minimize the project impact on local households.

The project has no trans-boundary environmental and social impacts.

The domestic law and regulation do not require any environmental impact assessment. According to applicable law, the project owner is required to conduct a pre-assessment where they need to prove that the project site is not particularly sensitive to any environmental and social impact. Such pre-assessment has been conducted and the project has been qualified not to have any further environmental and social assessment.

The project design identifies and addresses all social and environmental issues. The project pursues a preassessment report approved by the Ministry of Environment where all environmental and social concerns were identified through non-formal meetings and hearings with local stakeholders.

The headlines of the EIA Report prepared in 2010 are as follows:

KARADENİZ HES ELEKTRİK A.Ş. is planning to construct and operate Uzundere-I 63 MW HEPP near Cataldere village, in borders of Cayeli district of Rize province in Turkey.

Most of the project area is forestry lands and the company obtained all necessary permits for the use of these lands.

At project location, there is no irrigation activity and this water will be enough for natural river life. As the evaporation is very low, rainfall is high and there are many lateral rivers in the region, the water flow will be much higher than the water released from the weirs.

Some of the excavated earth were used as filling material at construction of the weir and the rest have been stored at storage areas.

The possible dust emission is calculated for the project and the results show that the emission amount is under the allowed limits in the “Regulation on Control of Industrial Air Pollution (dated 03.07.2009, numbered: 27277)”. In order to protect the air quality in the region, the construction activities were executed in line with the related laws and regulations.

Domestic, solid and liquid wastes are disposed in line with the regulations.

The limited damages to the environment arising from the project during the construction were minimized by taking all necessary precautions.

The project contributes to improve the environmental situation in the region and in the country as avoiding fossil fuel-based electricity will enhance the air quality and help to reduce the negative effects on the climate. Through renewable technologies and hydro-based electricity sustainable and climate friendly development is promoted. The project contributes to the Sustainable Development Goal, Climate Action.

During construction and operational period, the project has created employment opportunities for the local community. The project contributes the economic development of the region by providing sustainable energy resources. The project provides workers with a safe and healthy work environment and is not complicit in exposing workers to unsafe or unhealthy work environments. Thus, the project contributes to the Sustainable Development Goal, Decent Work and Economic Growth.

Regarding the wastes, hazardous wastes are handled appropriately in closed containers and transported by licensed transporters to the licensed processing and disposal facilities.

Wastewater is collected through within the septic tank and is transferred through these wage truck.

The domestic wastes are taken to the containers of the Municipality by the site personnel and the waste are transferred by the Municipality[3].

There is no agricultural area in the project area and the lands were owned by the state.

[3] The records are available to the DOE.

There is not any negative environmental or socio-economic impact.

2.4 Public Comments

Below is the list of stakeholder comments and the details of those comments were taken into account. The stakeholders addressed unemployment and weak economic activity as the major social problems in the area.

The local workers have been given priority during the hiring for plant operation. Also, during the construction, construction machines and equipment were hired from the vicinity of the project site. The project owner has procured all needs of the construction site from local resources in order to contribute to the local economy of the region. The project owner helped local people and the institutions in many ways, from renewing the electricity infrastructure to building roads in the forest to help forest workers in fighting against possible fires.

The second round of stakeholder consultation meeting within the scope of EIA which was ordered to be prepared by Rize Administrative Court was held in 14th April 2009. The attendants repeated the same comments that they did in previous meetings as briefed above.

2.5 AFOLU-Specific Safeguards

Uzundere I HEPP is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Project type: Type I – Renewable Energy Projects

Category: D – Electricity Generation for a System

Methodology: ACM0002: Grid-connected electricity generation from renewable sources, Version 20.0

Sectoral Scope: 01 Energy industries (renewable - / non-renewable sources)

ACM0002 refers to:

- Tool to calculate the emission factor for an electricity system, Version 07.0,
- Tool for the demonstration and assessment of additionality, Version 07.0.0
- Tool "Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period", Version 03.0.1

3.2 Applicability of Methodology

The methodology ACM0002: Grid-connected electricity generation from renewable sources is applicable to grid-connected renewable power generation project activities that a) install a Greenfield power plant; b) involve a capacity addition to (an) existing plant(s); c) involve a retrofit of (an) existing operating plants/units; d) involve a rehabilitation of (an) existing plant(s)/unit(s); or e) involve a replacement of (an) existing plant(s)/unit(s).

The project activity installs a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield), ACM0002: Grid-connected electricity generation from renewable sources is applicable. The applicability criteria are listed and justified below:

The choice of methodology ACM0002 Version 20.0 is justified as the proposed project activity meets relevant applicability criteria:

Table 3 - Applicability of ACM0002

Applicability Criteria	**Justification**
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or	The project is installation of a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity.

(e) Involve a replacement of (an) existing plant(s)/unit(s)	
The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity	The project is a hydro power plant.
In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or	The project activity results in new single reservoir and the power density, is greater than 4 W/m2.

(b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (3), is greater than 4 W/m2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (3), is greater than 4 W/m2. (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2, all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2 ; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project does not involve switching from fossil fuels to renewable energy sources and is not a biomass fired power plant.

In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project does not involve retrofits, rehabilitations, replacements, and it’s not a capacity addition.

Additionally, the proposed project activity meets applicability criteria of TOOL07: Tool to calculate the emission factor for an electricity system; Version 7.0:

- Para 3: This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).

  Project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid. There is only one national grid in Türkiye.

- Para 4: Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.

  The emission factor for only grid power plants (off grid power plants are not taken into account) have been used by Turkish Ministry of Energy and Natural Resources to calculate emission factor.

- Para 5: In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.

  Tool restricts use of Tool 07 to non-annex 1 countries but that is for CDM application, this project is voluntary reduction carbon project and thus can apply the Tool 07 in Türkiye (Annex-1) country. Specially Türkiye has special circumstances and did not take any place withing compliance market.

- Para 6: Under this tool, the value applied to the CO2 emission factor of biofuels is zero. The calculation of the emission factor does not involve any biofuels.

  The calculation of the emission factor does not involve any biofuels.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Generation mixes of electricity grid in Turkey	CO_2	Yes	Main source. The dominant emissions from power plants are in the form of CO2, therefore CO2 emissions from fossil fuel fired power plants connected to the grid will be accounted for in baseline calculations.
		CH_4	No	-
		N_2O	No	-
Project	Construction and operation of HEPP	CO_2	No	Not applicable

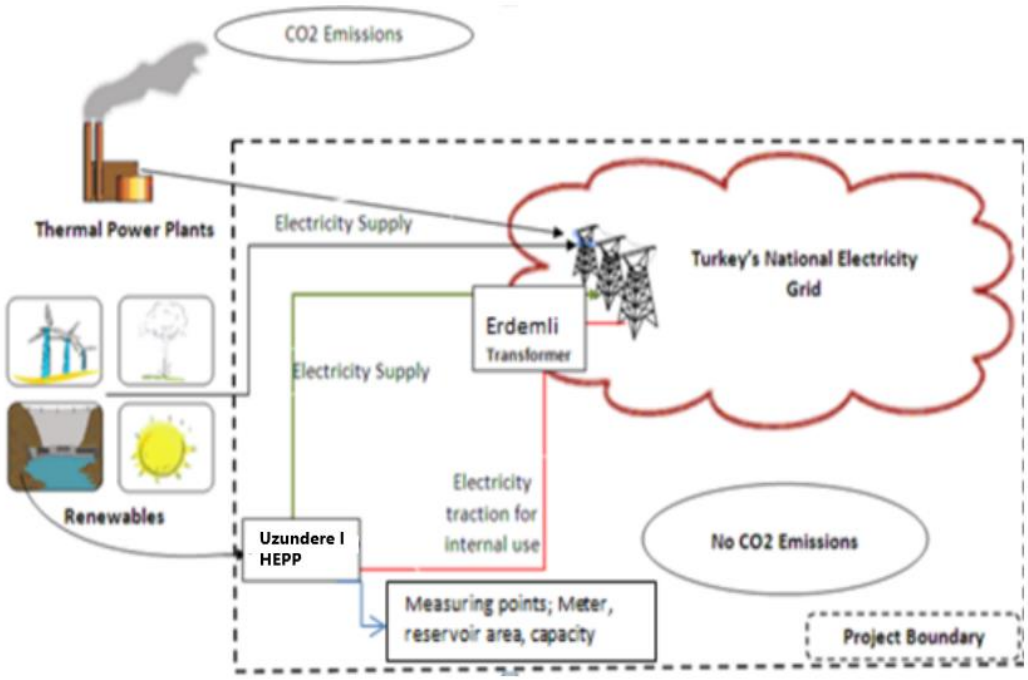

Figure 3. Project Boundary

3.4 Baseline Scenario

According to ACM0002 (Version 20), if the project activity is the installation of a new grid-connected renewable power plant, the baseline scenario is the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Step 1: Assess the validity of the current baseline for the next crediting period

The current baseline complies with all relevant mandatory national and/or sectoral policies which have come into effect after the submission of the project activity for validation or the submission of the previous request for renewal of the crediting period and are applicable at the time of requesting renewal of the crediting period.

The following applicable mandatory laws and regulations have been identified.

- Electricity Market Law[4]

  The project operated under the generation license which was given by the market operator (EMRA) and all licensees have to be act in line with this law.

- Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy

  The project utilizes hydro power and all renewable energy generators have to act in line with this law.

- Environmental Law[5] and Regulation on Environmental Impact Assessment (EIA)[6]

  The environmental indicators have been monitored and reported to the regulatory authorities in accordance with the EIA Regulation during 1st crediting period. The legal permissions and licenses have already been taken.

- Regulation on procedures and principles of signing the agreement of utilisation of water resources for the purpose of electricity production in the electricity market[7]

All hyro power generators have to sign and act in line this regulation. Otherwise, no activity is allowed.

Finally, the project is not mandated by any enforced law, statute or other regulatory framework, and that the energy generated by project activity is not used to meet governmental targets, laws, or legal mandates. The project operates within free market conditions.

Step 1.2: Assess the impact of circumstances

As it may be seen in Figure 4, the development of Turkey's installed capacity by primary energy resources between the years, 2009-2019, the electricity generation has mainly been done by fossil fuel fired power plants in Turkey. Total Installed electricity generation capacity in Turkey has reached 91,267 megawatts (MW) as of 2019. As having a share of 31.23%, wind power projects have an installed capacity of 28,503 MW.

[4] https://www.mevzuat.gov.tr/mevzuatmetin/1.5.6446.pdf

[5] https://www.mevzuat.gov.tr/mevzuatmetin/1.5.2872.pdf

[6] https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=39647&MevzuatTur=7&MevzuatTertip=5

[7] https://www.resmigazete.gov.tr/eskiler/2015/02/20150221-7.htm

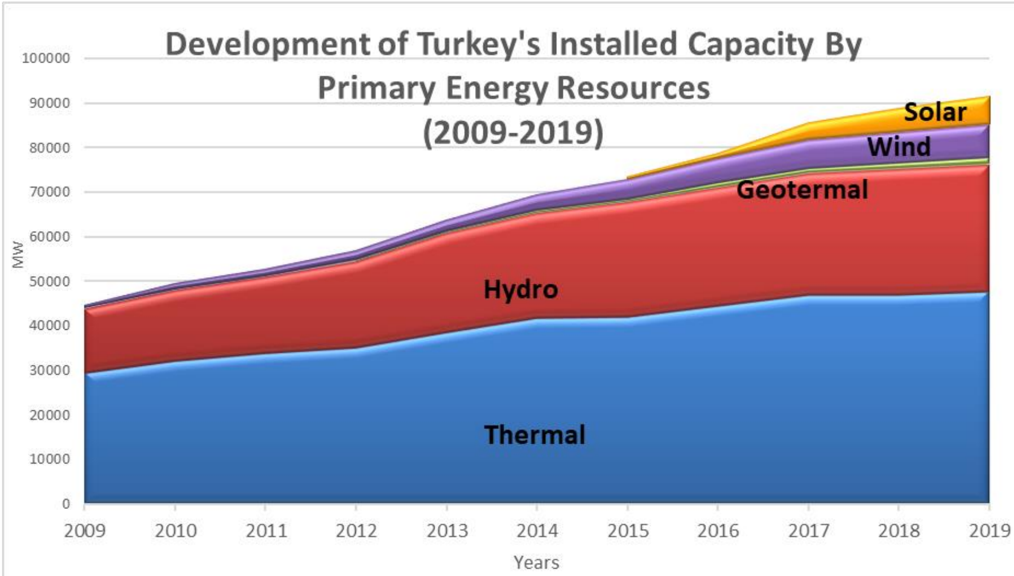

Figure 4. The development of Turkey's installed capacity by primary energy resources, 2009-2019

In reference to 5-year capacity projection[8], it is clear that fossil fuels will remain the main sources for electricity generation through until 2024. Fossil fuels will continue to dominate the market. Hydro will account for 15% of the mix whereas all non-hydro renewable combined (geothermal/ biomass/ solar/ wind) will only account for 11% of all electricity generation capacity. This projection is consistent with continuing fossil fuel dependent characteristics of Turkish electricity sector.

[8] https://webapi.teias.gov.tr/file/abeac87d-3abc-4532-9cf4-d6f3a9d34c17?download

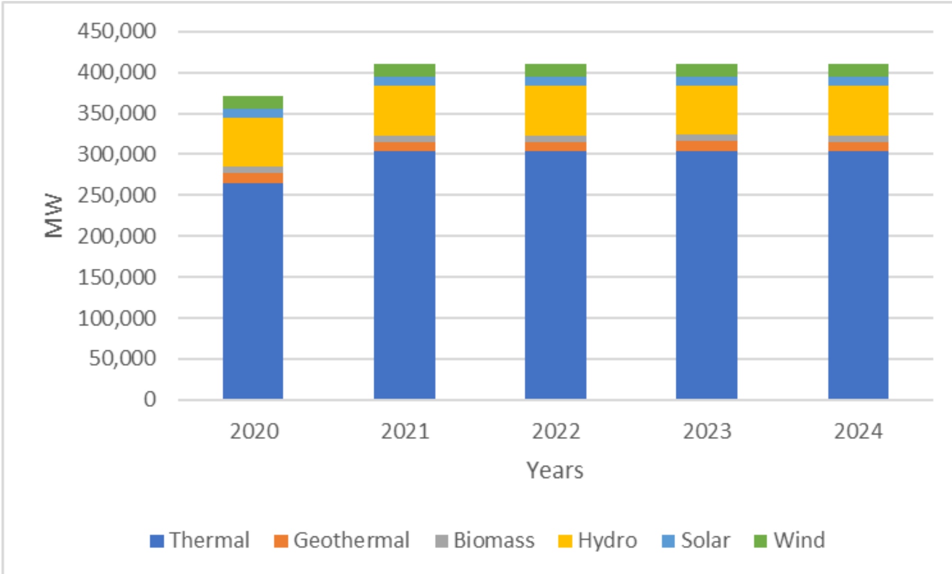

Figure 5. 5-year capacity projection

The current baseline has been updated with the latest data and projections available by the official bodies. It's clear that the baseline scenario is still valid for the second crediting period in accordance with the tool "Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period".

Step 1.3: Assess whether the continuation of use of current baseline equipment(s) or an investment is the most likely scenario for the crediting period for which renewal is requested

This sub-step is not applicable since the baseline scenario identified at the validation of the project activity was not the continuation of use of the current equipment(s) without any investment.

Step 1.4: Assessment of the validity of the data and parameters

Sections 4 and 5 have been updated.

The current baseline has been updated with the latest data and projections available by the official bodies. It's clear that the baseline scenario is still valid for the second crediting period.

3.5 Additionality

Demonstration and Assessment of Additionality Version 7.0.0" has been applied to the project. The additionality has been evaluated in first validation and that the information is repeated in this PDD and no new additionality assessment is done.

The project is not being implemented to comply with any law or other regulatory framework.

The IRR of the project was compared against an investment benchmark which is derived by the CAPM scientific model and was calculated as 15.65 %.

IRR Calculation of the Project

IRR Calculation of the project is based on the feasibility study prepared for the Board of the Company in 2007. Aiming to show that the income of the VERs is important for the financial performance of the project, two scenarios are presented in this section, one which excludes revenues from the sale of VERs and the second which includes the revenues from the sale of VERs. The assumptions used for this analysis are outlined as follows:

- The financial analysis was performed over the 35-year period from 2010 to 2045, this, therefore includes the investments made by the project owner between 2007 and 2010 and the operational costs along the lifetime of the project.
- A tax rate of 20% was applied to the project in line with Turkish tax laws.
- The depreciation period of machinery and equipment was assumed as 30 years as the local law allows companies to depreciate machinery and equipment in 30 to 40 years. In the meantime, the other fixed assets" (buildings etc.) depreciation life was assumed as 40 years.
- The annual power generation figure has been 156,205,000 kWh.
- The power purchase price for the project was assumed to be 7.49$ cent per kWh which is calculated by converting the 5.5 Euro cent using the monthly average exchange rate (15th August 2007 – 14th September 2007) before the date board decision regarding the investment dated 14th September 2007.)
- The revenues from VERs are excluded from scenario 1 and included in scenario 2. The volume of VERs generated by the project is calculated by multiplying the annual electricity output of the project by the emission factor. For sensitivity analysis, the revenues related to the sale of the VERs are applied by multiplying the volume of VCUs by the $5, $6 and $7 price.

Sensitivity analysis

Sensitivity analysis was applied to Investment Costs, Energy Production and Electricity Sales Price. Applying variations on investment costs, the range was in a bandwidth of +10% and –10%.

The sensitivity in the investment amount was applied to the investment items which represent equal or above 20% of the total investment amount.

Table 4 -Sensitivity to Variation in Investment Costs

Investment Costs	**-10,0%**	**-5,0%**	**0,0%**	**5,0%**	**10,0%**
Without VER	8,68%	8,42%	8,16%	7,92%	7,69%
IRR @5 USD/ton VER	9,02%	8,74%	8,47%	8,22%	7,97%
IRR @6 USD/ton VER	9,08%	8,80%	8,54%	8,28%	8,03%
IRR @7 USD/ton VER	9,15%	8,87%	8,60%	8,34%	8,09%

Variation on energy production was also applied in a bandwidth of +10% and –10% on the secondary energy production which reflects the volatility in calculations. Please note that “primary” part of the production reflects the electricity production which is 90% guaranteed amount. Therefore, no sensitivity was applied to this part.

Table 5 - Sensitivity to Variation in Electricity Production

Electricity Production	**-10,0%**	**-5,0%**	**0,0%**	**5,0%**	**10,0%**
Without VER	6,72%	7,44%	8,16%	8,89%	9,63%
IRR @5 Eur/ton VER	6,97%	7,72%	8,47%	9,23%	10,01%
IRR @6 Eur/ton VER	7,03%	7,78%	8,54%	9,30%	10,09%
IRR @7 Eur/ton VER	7,08%	7,83%	8,60%	9,38%	10,17%

Due to the fact that the feed in guaranteed tariff prices is fixed in Turkey, the unit energy price is not considered to be an exogenous variable. However, sensitivity on the upward side was also applied to see the effect of price changes.

Table 6 - Summary IRRs for Different VER prices

Electricity Price	**-10,0%**	**-5,0%**	**0,0%**	**5,0%**	**10,0%**
Without VER	N/A	N/A	8,16%	8,89%	9,63%

IRR @5 Eur/ton VER	N/A	N/A	8,47%	9,22%	9,98%
IRR @6 Eur/ton VER	N/A	N/A	8,54%	9,28%	10,05%
IRR @7 Eur/ton VER	N/A	N/A	8,60%	9,35%	10,12%

The above sensitivity analysis reveals the fact that no alternative scenario without VER revenues can make the project pass the benchmark IRR expectation which is 15.65% . Therefore, the project is not financially attractive

Common Practice Analysis

The common practice analysis was done first validation and that the information is not repeated in this PDD and no new common practice analysis has been done.

3.6 Methodology Deviations

There are no methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Since the project is an installation of a new grid-connected renewable power plant, the baseline scenario is formulated in ACM0002, Version 20: “Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the Combined Margin (CM) calculations described below”.

The baseline emission factor is the weighted average of the Operating Margin Emission Factor and Build Margin Emission Factor.

The combined margin (CM) emission factor of Turkish National Transmission System has been published by the Ministry of Energy. The latest values belong to the 2018 statistics[9].

[9] https://enerji.gov.tr/duyuru-detay?id=82

Calculation of the Operating Margin Emission Factor

It's been published as 0.7258 tCO2/MWh by the Ministry of Energy.

Calculation of the Build Margin Emission Factor
It's been published as 0.4153 tCO2/MWh by the Ministry of Energy.

Calculating of the Combined Margin Emission Factor
It's been published as 0.4929 tCO2/MWh by the Ministry of Energy.

Therefore resulting EFgrid,CM,y is 0. 0.4929tCO2 /MWh.

Accordingly, the baseline emissions BEy are calculated as following:

$$BEy = EGy \times EFgrid,\ CM,\ y$$

Where:

BE_y	: Baseline emissions (tCO2e)
EG_y	: Annual electricity supplied by the project to the grid (MWh)
$EF_{grid,\ CM,\ y}$	: Baseline emission factor (tCO2e/MWh)
y	: Refers to a given year

Also, as suggested in ACM0002 / Version 20, the leakage emissions are not considered and assumed as "0".

4.2 Project Emissions

There is no project emission resulting from the reservoir area of the Project Activity as the power density of the project is greater than 10W/m2. Accordingly, The Project Activity's power density was calculated as below:

=63,000,000 W/ 4,901.72m2 [10]

= 12,852.6 W/m2

[10] The screenshot of the reservoir area

Since the power density is still greater than 10W/m2, CAP_{PJ} and A_{PJ} have not been put as monitored parameters during the second crediting period,

4.3 Leakage

Leakage emission (LEy) is considered as "0" as suggested in ACM0002, Version 20.

4.4 Net GHG Emission Reductions and Removals

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
27.05.2020-31.12.2020	46,198	0	0	46,198
2021	76,997	0	0	76,997
2022	76,997	0	0	76,997
2023	76,997	0	0	76,997
2024	76,997	0	0	76,997
2025	76,997	0	0	76,997
2026	76,997	0	0	76,997
2027	76,997	0	0	76,997
2028	76,997	0	0	76,997

2029	76,997	0	0	76,997
01.01.2030-26.05.2030	30,798	0	0	30,798
Total	769,969	0	0	769,969

Emission reductions are calculated as follows:

$ERy = BEy - PEy\text{-}\ LEy$

Where:

ERy	Emission reductions in year y (t CO2e/yr)
BE_y	Baseline emissions in year y (tCO_2e/y)
PEy	Project emissions in year y (t CO2e/yr)
LEy	Leakage emissions in year y (t CO2e/yr)

ERy = 76,997-0-0

ERy = 76,997tCO2e/y

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EFgrid,CMy
Data unit	tCO2/MWh
Description	Emission factor of the Turkish grid determined ex-ante. It's been published by the Ministry of Energy for 2019.
Source of data	Ministry of Energy. Please see: https://enerji.enerji.gov.tr/Media/Dizin/BHIM/tr/Duyurular/ /Bilgi_Formu_Web_Sitesi_2019_202110071443.pdf
Value applied	0.4929
Justification of choice of data or description of measurement methods and procedures applied	Emission factor of the Turkish grid published by the Ministry of Energy for 2019.
Purpose of Data	Calculation of the baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	EPIAS records
Description of measurement methods and procedures to be applied	A set of meters measuring continuously then calculate the net electricity generation supplied by the project to grid.
Frequency of monitoring/recording	Measuring continuously/ recording monthly
Value applied	The annual electricity fed to the grid is estimated as 156,205 MWh.
Monitoring equipment	The power generated and supplied is monitored continuously and is recorded daily and aggregated into monthly totals. The

<table>
<tr><td></td><td>meters are both owned and installed by TEIAS (Turkish Electricity Transmission Company). The meters are sealed also by TEIAS. The meters are firsthand, and the supplier company commits that these meters fully conform or exceeds all relevant IEC standards including those dealing with electronic metering equipment IEC61036 for class 1 equipment and IEC60687 for class 1S equipment. There is no possibility for human error in the measurement of the electricity. All the measurements and calculations are done via tested meters.

Meters information:
<table><tr><th>Specifications</th><th>Main meter</th><th>Spare Meter</th></tr><tr><td>Manufacturer:</td><td>EMH</td><td>EMH</td></tr><tr><td>Serial No:</td><td>9674605</td><td>9674606</td></tr><tr><td>Accuracy Class:</td><td>1S</td><td>1S</td></tr></table>
The following meters were changed on 15.10.2020:
<table><tr><th>Specifications</th><th>Main meter</th><th>Spare Meter</th></tr><tr><td>Manufacturer:</td><td>ACTARIS</td><td>ACTARIS</td></tr><tr><td>Serial No:</td><td>53064430</td><td>53064429</td></tr><tr><td>Accuracy Class:</td><td>0.5S</td><td>0.5S</td></tr></table>
The installation and latest test date of the meters are 15.10.2020[11].</td></tr>
<tr><td>QA/QC procedures to be applied</td><td>The OSF forms have been used to cross-check.
Calibration of all the meters will be undertaken at required intervals and faulty meters will be duly replaced immediately.</td></tr>
<tr><td>Purpose of data</td><td>Calculation of emission reductions</td></tr>
<tr><td>Calculation method</td><td>$EG_{Facility,y} = EG_{export,y} - EG_{import,y}$</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

5.3 Monitoring Plan

[11] Meter tests are available to the DOE.

The Project Owner will be responsible for the overall management of the monitoring procedures including recording, data collection and store. The consultant will calculate emission reductions based on these monitored data and prepare monitoring report.

According to the methodology applied, the electricity supplied to the national grid by the project and the electricity consumed by the project activity shall be monitored. The net electricity is the difference of the electricity supplied and consumed by the project and shall be taken into account for emission reduction calculations.

Two power meters are installed at the grid interface of the project. One of them is the main meter and the other one is back-up meter of the main meter for cross-checking. Both meters are jointly inspected and sealed in order to be protected from interference by any of the parties.

The capacity of the transmission line connected is 154 kVA, the accuracy class for main power meters have been defined in the Communiqué for Power Meters as 1S class. The back-up meter has the same accuracy class of 1S. The calibration will be implemented in accordance with the related standard procedures (IEC-EN 62053-22 and 62053-23) by either Turkish Electricity Transmission Corporation (TEIAS) or the provider company in the name of TEIAS. The meters are calibrated on yearly basis. The latest calibration was made on 15/10/2020.

TEİAŞ is performing remote reading of the meters and monthly power meter readings are the basis for monitoring net electricity fed into the grid. EPİAŞ records will used as the source of net generated electricity value and meter reading forms or OSF forms issued by TEİAŞ will be used for the crosscheck.

The website of EPİAŞ (https://cas.epias.com.tr/cas/login) is accessible to Project owner with their unique user ID and password. Once accessed, the Project owner is able to call electricity generation and consumption reports of their own projects. The same reports are used by the Project owner for invoicing TEİAŞ. The electricity generation data is reported monthly basis.

All data collected (especially, EPIAS records and OSF Forms) as part of monitoring will be archived electronically by the project owner and be kept at least for 2 years after the end of the last crediting period.

Since the installed capacity (CapPJ) of the project activity and the reservoir area (APJ) have not been changed during the 1st crediting period, these parameters have been removed from Data and Parameters Monitored.